# Andrew W. Shenstone

Shenstone, Andrew - Application for Software Engineer — Activation @ Vercel

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## SUMMARY

Senior engineer with 12+ years; strongest in TypeScript, JavaScript, and Next.js. Plantfolio runs on Vercel, so I spend every day in the same kind of stack your customers use. As a founder I talked with users to find drop-off, tightened onboarding, and shipped changes where we could see whether more people finished the job—not just whether we shipped tickets. I start from a clear question, put basic measurement in place, and iterate. I use Flutter (Dart) for mobile work alongside React Native. Go is not a language I have leaned on professionally; if this role needs Go, I will put real time into learning it. I have not had the title growth engineer and I have not been the main owner of a company-wide A/B testing platform; that is work I want to grow into here.

## SKILLS

**Platform & frontend:** TypeScript, JavaScript, Next.js, React, Vercel, Node.js, Fast pages and healthy Core Web Vitals

**Product-led engineering:** Customer discovery & friction triage, Hypothesis-driven iteration, End-to-end ownership: prototype through production, Quality: Vitest, Cypress, code review

**AI/ML:** OpenAI API, AI SDK, Vector Embeddings, Natural Language Processing, Retrieval Augmented Generation (RAG)

**Architecture & async:** Monorepos, Async workflows (Inngest, webhooks, background jobs), ETL pipelines, Performance optimization

**Cloud & Tools:** AWS (S3, EC2), Vercel, Algolia, Inngest, Clerk Auth, Stripe, Git, CI/CD, pnpm, Turbo, Cursor IDE (Top 1% user)

## EDUCATION

B.S. Finance, Santa Clara University — 2010

## EXPERIENCE

### Founder & Lead Engineer

**Plantfolio** | November 2023 - Present

#### Product Leadership & Architecture

- Architected and built a comprehensive plant management platform from the ground up, serving as both technical lead and product owner
- Owned the full feature cycle: joined customer calls to identify friction points, architected solutions, implemented, tested, and verified success in production
- Designed and implemented a sophisticated monorepo architecture with 12+ packages, enabling rapid development and maintainable codebases
- Led product strategy and user experience decisions, making product micro-decisions and navigating ambiguity to deliver value iteratively

#### Technical Achievements

- AI Integration: Built advanced AI-powered features including plant identification from photos, contextual chatbot (Sage), and smart text extraction from plant tags using OpenAI's vision and language models
- Modern Frontend Architecture: Implemented Next.js with React, React Server Components, and TypeScript, achieving exceptional performance and developer experience
- Complex Data Modeling: Designed and implemented sophisticated PostgreSQL schemas with Prisma, handling complex relationships between users, plants, folios, and media with 15+ interconnected models
- Supabase to Neon: Shipped on Supabase (PostgreSQL with Row-Level Security policies for multi-tenant isolation) before migrating the data layer to Neon; strong hands-on experience with RLS and Postgres-centric access control

- Content Management: Created sophisticated content management system with rich text editing, plant tagging, and flexible data structures

### Scale & Performance

- Managed a database of 40,000+ plant species with advanced search capabilities using Algolia and vector embeddings
- Implemented comprehensive media management system with AWS S3, supporting image optimization, galleries, and metadata extraction
- Built robust authentication and authorization systems with Clerk, supporting multiple user types and subscription tiers
- Optimized application performance through React Server Components, efficient data fetching, and strategic caching

### Client Case Study - AI Signage

- Delivered a production rollout for an independent garden center to automate plant signage generation; one independent retail customer in ongoing production who has been an enthusiastic reference
- Enabled bulk plant import (plain text) to printable signs in seconds; reduced a weekend-long manual process to minutes after initial setup
- Demonstrated ability to translate customer requirements into scalable technical solutions and shorten feedback loops

### Full-Stack Ownership

- Developed complete payment integration with Stripe, subscription management, and user onboarding flows
- Built comprehensive email system with React Email, supporting transactional emails, notifications, and marketing campaigns
- Implemented advanced search and filtering capabilities with Algolia, supporting complex plant queries and user preferences

### Code Quality & Leadership

- Established coding standards and architectural patterns that enabled rapid, consistent development

- Implemented comprehensive testing strategy with Vitest and Cypress, maintaining high code quality
- Built reusable component library and design system, ensuring consistent UI/UX across the platform

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## Senior Software Engineer

**Tilia** | January 2023 - August 2023

- Partnered with cross-functional teams in product and design to revamp critical user-facing features
- Built and shipped work in support of KYC, identity verification, and risk-adjacent tooling as part of Tilia's payments platform (tools team context)
- Spearheaded and guided a team through the successful adoption of Chakra UI, reducing bespoke components and establishing cohesive design patterns
- Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
- Maintained exceptional unit test coverage of over 90% for the Tools team application, alongside comprehensive end-to-end testing
- Led the swift creation and deployment of a homepage reporting dashboard within the initial two months of joining
- Role ended in a company-wide layoff affecting approximately one-third of employees

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## Senior Software Engineer

**Bloom.wine** | September 2020 - October 2022

- Architected a greenfield Next.js application for the primary admin interface of the Bloom product; much of that stack remains in production across many winery clients today

- Engineered Ruby on Rails GraphQL API endpoints shared by multiple applications across mobile and web devices
- Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
- Collaborated with designers using Figma, establishing component library standards usable by both engineering and design
- Integrated DatoCMS and extended application functionality through third-party API integrations (Shopify, Mailchimp, Stripe, Shipstation, ActiveCampaign); all Bloom winery clients were on Shopify Plus

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## Software Engineer II

**Tune Inc - Marketplace Automation** | August 2019 - May 2020

- Built and operated ETL-style data pipelines and integrations supporting marketplace automation workflows
- Automated a client workflow using a Node.js/Express API and React.js, reducing the time to complete the task from 10-15 minutes to 10-30 seconds
- Modernized the primary legacy React.js application by moving from a custom Webpack process to react-scripts, resulting in a 50% reduction in production build size and a 60% reduction in NPM dependencies
- Reduced development machine CPU usage from 75% to <5% by simplifying the build pipeline and dependency graph
- Role ended in a company-wide layoff

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## Software Developer

**Ply Interactive Inc** | June 2014 - July 2019

### React Native

- Shipped multiple production React Native applications for agency clients—cross-platform mobile apps delivered end to end

- Washington State Fair: React Native app in sustained multi-year production; very high traffic during fair season, from tens of thousands of users into the hundreds of thousands cumulatively over seasons

### Web & platforms

- Developed numerous web applications including Georgetownbeer.com, WashingtonWine.org, TwoWolvesWine.com, Novaerus.com, WellAir, and Plasma-air.com
- Built a standalone Ruby gem Content Management System, enabling simplified content management within Ruby on Rails projects
- Transitioned the agency to modern technologies like GraphQL, Gatsby.js, and React.js, improving delivery times for new projects
- Bloom.wine Product Development: Built extensive reporting features using complex PostgreSQL queries and interactive filtering, saving clients thousands of hours on previously manual tasks
- Engineered a flexible Wine Club membership structure for wineries, allowing for diverse client configuration and broadening the range of potential sales targets
- Communicated and enforced the standards of Shopify's Polaris React.js component library with designers, resulting in efficiencies gained from leveraging pre-built components

## COMPANY REFERENCE

**Plantfolio** (2023-Present) - AI-powered plant management platform with 40,000+ plant database serving both consumer garden management and independent garden center services - plantfolio.garden/

**Tilia** (2023) - All-in-one payment platform for digital economies - tilia.io/

**Bloom.wine** (2020-2022) - Commerce services for the alcohol industry; all clients on Shopify Plus - bloom.wine/

**Tune Inc - Marketplace Automation** (2019-2020) - SaaS platform for managing marketing partnerships across mobile and web - tune.com/

**Ply Interactive Inc** (2014-2019) - Boutique agency: web and production React Native apps (e.g. Washington State Fair at major scale); developed Bloom.wine product internally (2017-2019)